

Solution Requirements (Functional & Non-functional)

Date: 30 September 2026

Team ID: Msu207Team5 6ab24d09e6af97f73bc88d13

Project Name: Import Data Using Transform Maps in Sheets

Maximum Marks: 4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data File Preparation	Prepare data in Google Sheets Prepare data in CSV format Maintain required column names Check data before uploading
FR-2	Data Upload	Upload the Sheet/CSV file Read source data Check file format Display uploaded records
FR-3	Transform Map Configuration	Create Transform Map Select source table Select target table Configure import settings
FR-4	Field Mapping	Map source fields to target fields Review field mappings Modify incorrect mappings Map required fields
FR-5	Data Transformation	Apply transformation rules Convert source data into target format Modify field values Handle different data formats
FR-6	Data Validation	Validate required fields Check missing values Check invalid data Verify data format before import

FR-7	Data Import	Import valid records Insert records into target table Process multiple records Complete automated data transfer
FR-8	Error Handling	Identify failed records Display import errors Correct invalid records Reprocess failed records
FR-9	Duplicate Handling	Identify duplicate records Compare existing records Prevent unnecessary duplicate entries Maintain data consistency
FR-10	Import Monitoring	View import status Check successful records Check failed records View import summary
FR-11	Data Verification	Verify imported records Check mapped field values Compare source and target data Confirm successful data storage
FR-12	Transform Map Management	Edit mapping configuration Update transformation rules Save mapping changes Manage different import requirements

Non-functional Requirements

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide a simple and easy-to-understand process for uploading files, mapping fields, validating data, and importing records. Users should be able to perform the import process without unnecessary complexity.
NFR-2	Security	The system should protect imported data from unauthorized access. Only authorized users should be allowed to configure

		Transform Maps, upload data, and perform administrative operations.
NFR-3	Reliability	The system should perform the data import process consistently and accurately. It should handle errors properly and avoid losing valid records during the import process.
NFR-4	Performance	The system should process and import data within a reasonable time. It should provide good performance even when the uploaded file contains a large number of records.
NFR-5	Availability	The system should be available whenever users need to upload and import data. The import functionality should remain accessible during normal working hours with minimum service interruption.
NFR-6	Scalability	The system should be able to handle an increasing number of records and larger data files. It should support future growth without requiring major changes to the complete system.